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```
===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
3. Display Devices
4. Display Adjacency Matrix
5. Display Device Connections
6. Explore Devices Using DFS
7. Exit
Enter your choice: 1
Enter device name: smart gateway
Device added with ID: 0

===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
3. Display Devices
4. Display Adjacency Matrix
5. Display Device Connections
6. Explore Devices Using DFS
7. Exit
Enter your choice: 1
Enter device name: smart light
Device added with ID: 1

===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
3. Display Devices
4. Display Adjacency Matrix
5. Display Device Connections
6. Explore Devices Using DFS
7. Exit
Enter your choice: 1
Enter device name: securaty camera
Device added with ID: 2

===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
```

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```
===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
3. Display Devices
4. Display Adjacency Matrix
5. Display Device Connections
6. Explore Devices Using DFS
7. Exit
Enter your choice: 2

--- IoT Devices ---
0 : smart gateway
1 : smart light
2 : securaty camera
Enter first device ID: 0
Enter second device ID: 2
Devices connected successfully!

===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
3. Display Devices
4. Display Adjacency Matrix
5. Display Device Connections
6. Explore Devices Using DFS
7. Exit
Enter your choice: 2

--- IoT Devices ---
0 : smart gateway
1 : smart light
2 : securaty camera
Enter first device ID: 1
Enter second device ID: 3
Invalid device IDs!

===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
```

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```
===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
3. Display Devices
4. Display Adjacency Matrix
5. Display Device Connections
6. Explore Devices Using DFS
7. Exit
Enter your choice: 2

--- IoT Devices ---
0 : smart gateway
1 : smart light
2 : securaty camera
Enter first device ID: 5
Enter second device ID: 0
Invalid device IDs!

===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
3. Display Devices
4. Display Adjacency Matrix
5. Display Device Connections
6. Explore Devices Using DFS
7. Exit
Enter your choice: 3

--- IoT Devices ---
0 : smart gateway
1 : smart light
2 : securaty camera

===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
3. Display Devices
4. Display Adjacency Matrix
5. Display Device Connections
```

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```
4. Display Adjacency Matrix
5. Display Device Connections
6. Explore Devices Using DFS
7. Exit
Enter your choice: 4

Adjacency Matrix:
   0 1 2
0: 0 0 1
1: 0 0 0
2: 1 0 0
1 = connected, 0 = not connected

===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
3. Display Devices
4. Display Adjacency Matrix
5. Display Device Connections
6. Explore Devices Using DFS
7. Exit
Enter your choice: 4

Adjacency Matrix:
   0 1 2
0: 0 0 1
1: 0 0 0
2: 1 0 0
1 = connected, 0 = not connected

===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
3. Display Devices
4. Display Adjacency Matrix
5. Display Device Connections
6. Explore Devices Using DFS
7. Exit
Enter your choice: 5
```

```
--- IoT Devices ---
0 : smart gateway
1 : smart light
2 : securaty camera
Enter device ID to view connections: 6
Invalid device ID!

===== IoT DEVICE NETWORK =====
1. Add Device
2. Connect Devices
3. Display Devices
4. Display Adjacency Matrix
5. Display Device Connections
6. Explore Devices Using DFS
7. Exit
Enter your choice: 7
Program terminated.

--------------------------------
Process exited after 117.2 seconds with return value 0
Press any key to continue . . .
```